Atmospheric Boundary Layer PDE Closure Benchmark Report

Empirical Discovery, Stability Diagnostics, and Galerkin PDE Benchmarks

AtmosphericSlowManifold.jl Pipeline

August 8, 2026

Abstract

This report synthesizes field campaign diagnostics (CASES-99, FLOSS, BLLAST, and SHEBA) and automated equation discovery for atmospheric boundary layer (ABL) turbulence parameterization. We present vertical eddy diffusivity $K_m(z)$ Bayesian credibility ribbons, Richardson number $Ri_g(z,t)$ heatmaps, and Pareto-optimal PDE closure benchmarks solved via spectral Galerkin discretization.

Contents

1 Theoretical Framework & Discovery Pipeline

Figure ?? illustrates the fast gravity wave phase-space decay onto the slow manifold $\mathcal{M}$, while Figure ?? outlines the equation discovery loop.

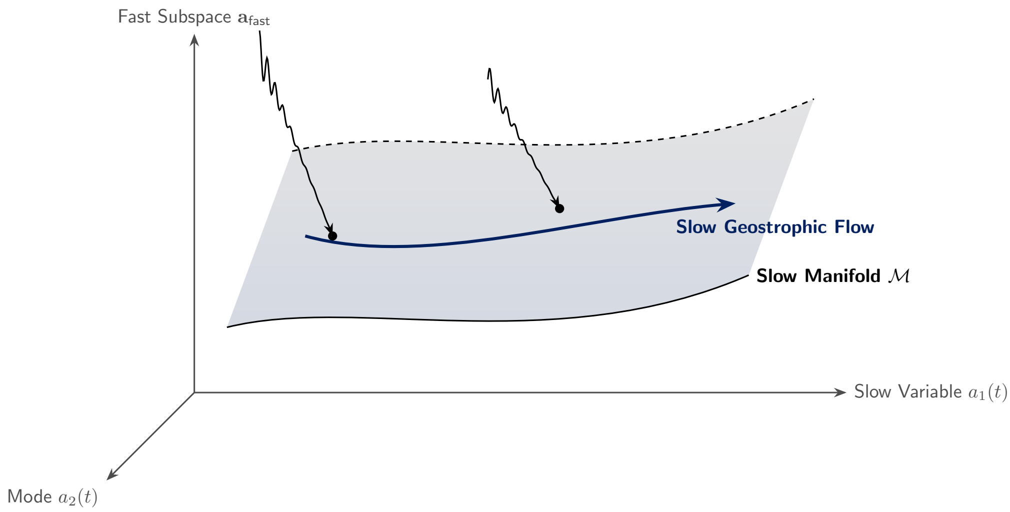


Figure 1: Phase-space collapse onto the lower-dimensional slow manifold $\mathcal{M}$ during spectral Galerkin integration.

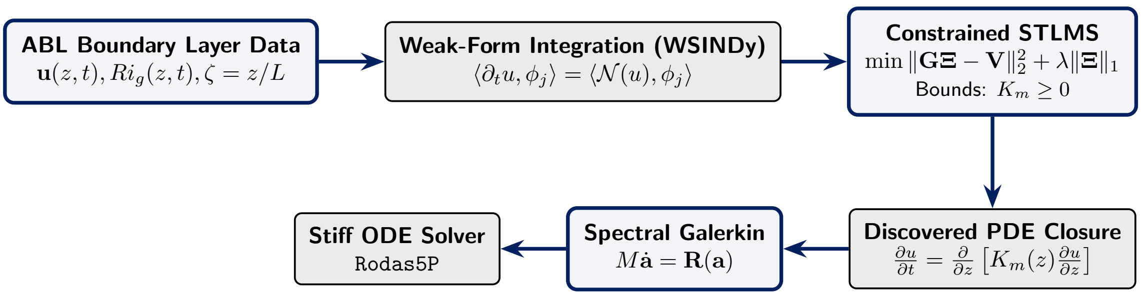


Figure 2: End-to-end weak-form SINDy discovery, constrained sparse optimization, and Galerkin PDE integration pipeline.

2 Field Campaign Dataset Diagnostics

Field campaign observations were ingested to derive Obukhov length L, stability parameter $\zeta = z/L$, non-dimensional wind shear $\phi_m(\zeta)$, and gradient Richardson numbers Ri_g.

Table 1: Field campaign dataset diagnostics, height levels, and stability parameter ranges.

Campaign	Observations	Height Levels	Mean Wind Speed ($\mathrm{m\,s^{-1}}$)	Median Richardson No
BLLAST	5600	18	11.335	0.172
SHEBA	2273	0	4.978	0.104
FLOSS	70796	18	28.755	0.225
CASES-99	6538	18	8.41	0.2

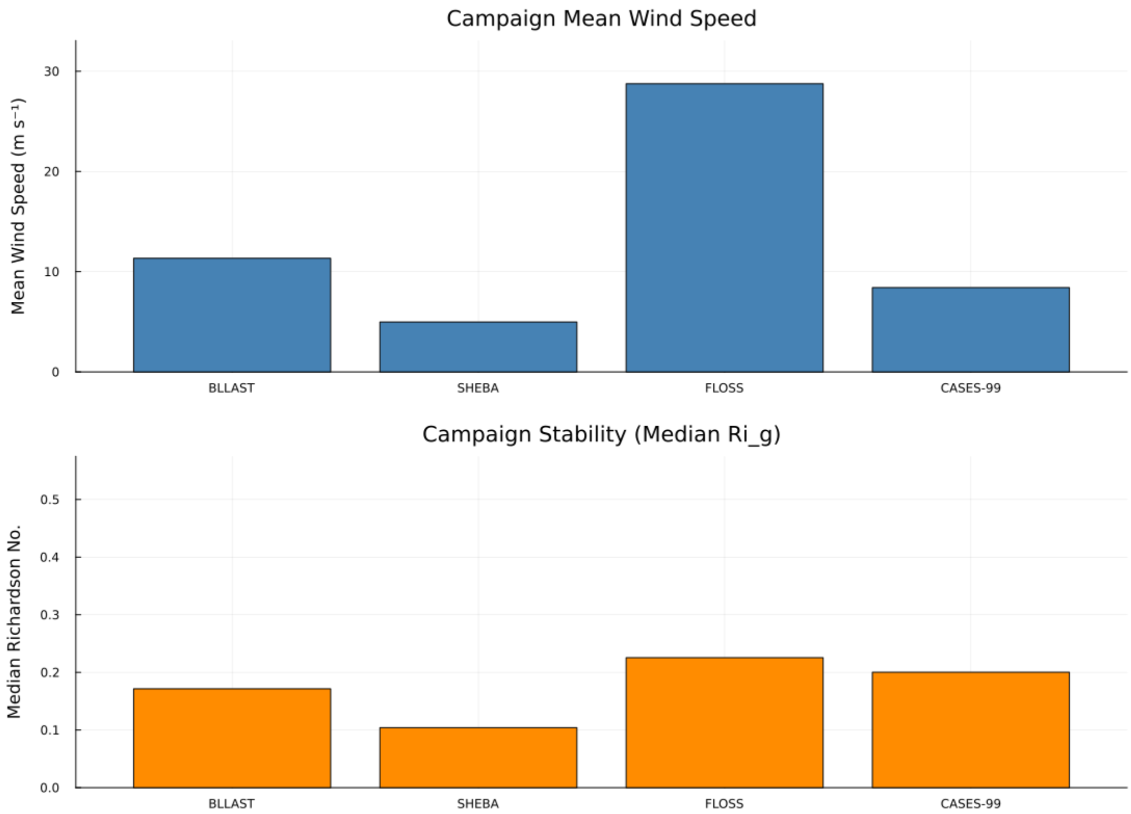
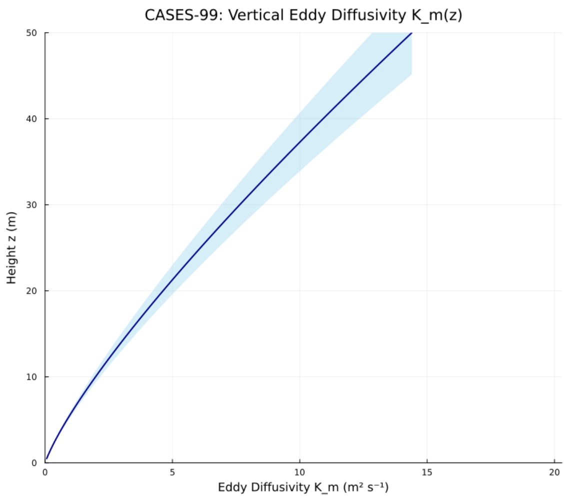


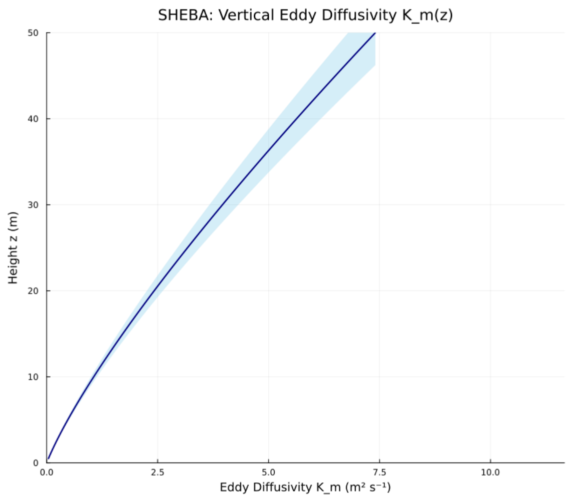
Figure 3: Comparative campaign overview displaying mean wind speed and stability metric profiles across all field sites.

2.1 Vertical Eddy Diffusivity $K_m(z)$ Uncertainty

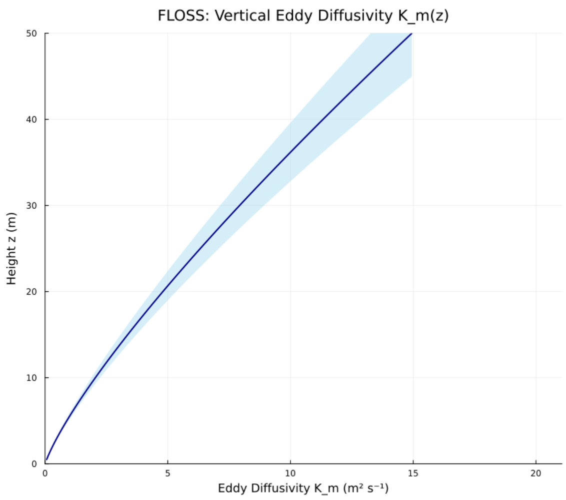
Bayesian parameter sampling propagates uncertainty through Similarity Theory formulations to construct 95% credibility ribbons for $K_m(z)$.



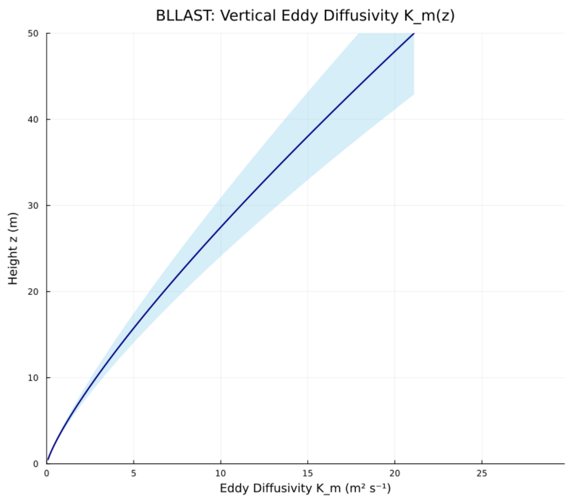
(a) CASES-99 $K_m(z)$ 95% Ribbon



(b) SHEBA $K_m(z)$ 95% Ribbon



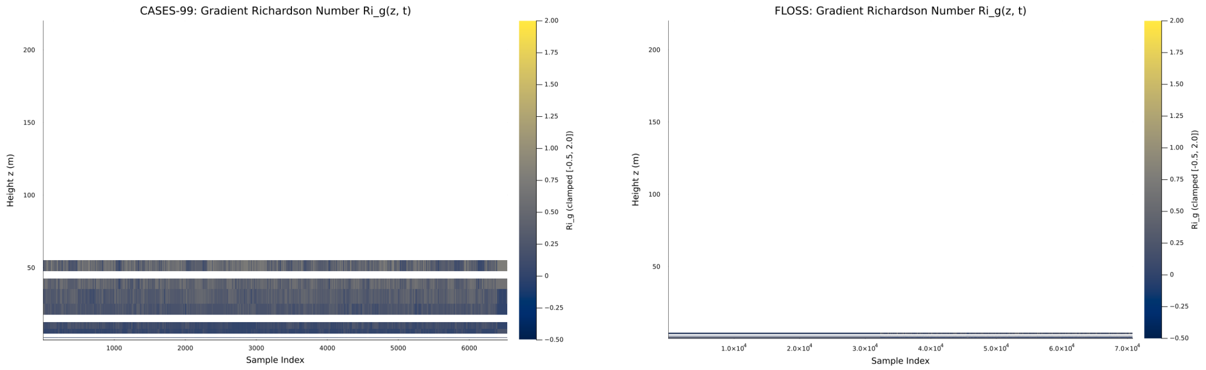
(c) FLOSS $K_m(z)$ 95% Ribbon



(d) BLLAST $K_m(z)$ 95% Ribbon

Figure 4: Vertical eddy diffusivity $K_m(z)$ profiles with median predictions and 95% posterior credibility intervals across target field campaigns.

2.2 Stability and Diagnostic Metrics



(a) CASES-99 $Ri_g(z, t)$ Heatmap

(b) FLOSS $Ri_g(z, t)$ Heatmap

Figure 5: Gradient Richardson number time-height cross sections showing dynamic stability transitions.

3 PDE Closure Benchmark & Model Discovery

Discovered PDE closures are benchmarked against neutral baseline formulations using a 12-mode Gegenbauer Spectral Galerkin discretization integrated with stiff Rosenbrock/BDF ODE solvers.

3.1 Cross-Campaign Discovered Model Summary

Table 2: Cross-campaign Pareto-optimal discovered model performance metrics.

Site	Active terms	R^2	Residual norm
sheba	1	0.9966	0.27542
cases_99	1	-0.2289	3.19695

3.2 Leave-One-Site-Out (LOSO) Cross-Validation

To assess cross-campaign generalization, we perform Leave-One-Site-Out validation using campaign artifacts as observational inputs. Reported metrics summarize out-of-sample fit quality and a coefficient-consistency stability score across held-out sites.

Table 3: LOSO out-of-sample metrics by held-out site, with aggregate stability diagnostics.

Held-Out Site	Validation RMSE	Validation MAE	R^2	95% Coverage
SHEBA	1.1912	0.9547	0.0497	94.9%
CASES-99	3.9867	1.6645	-10.2371	99.6%
FLOSS	35.1974	15.4867	0.0661	92.8%
BLLAST	2.2804	2.0911	-0.4682	43.5%
Mean Validation RMSE: 10.6639	**Stability Score**: 1.000			

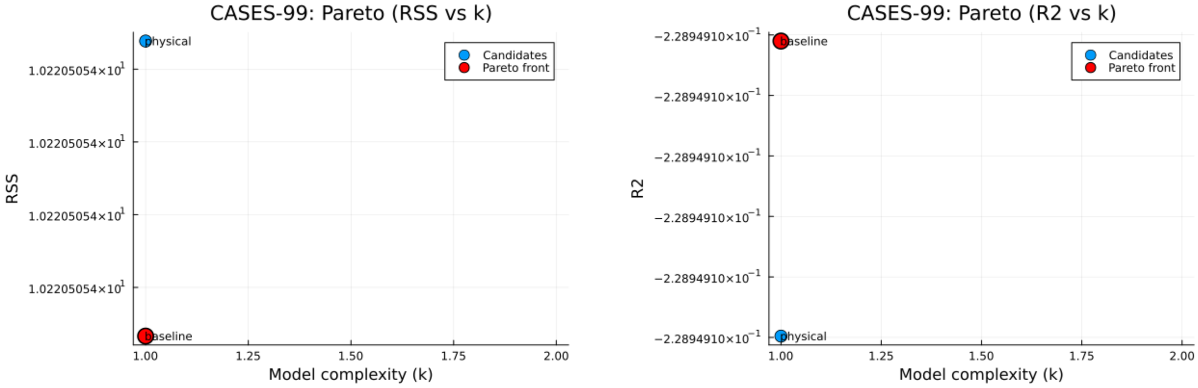
3.3 CASES-99 Optimal Pareto Closure

The optimal discovered differential equation governing momentum profile evolution for CASES-99 is:

$$\frac{\partial u}{\partial t} = 6.12 \, \frac{\partial^2 u}{\partial z^2}$$

Table 4: Discovered library terms and estimated coefficients for CASES-99.

Term	Basis	Coefficient
1	$\frac{\partial^2 u}{\partial z^2}$	6.12
$R^2 = -0.228949$		
Residual norm = 3.19695		



(a) CASES-99 Complexity vs RSS

(b) CASES-99 Complexity vs R^2

Figure 6: Pareto trade-off fronts between closure sparsity (k) and residual metrics for CASES-99.

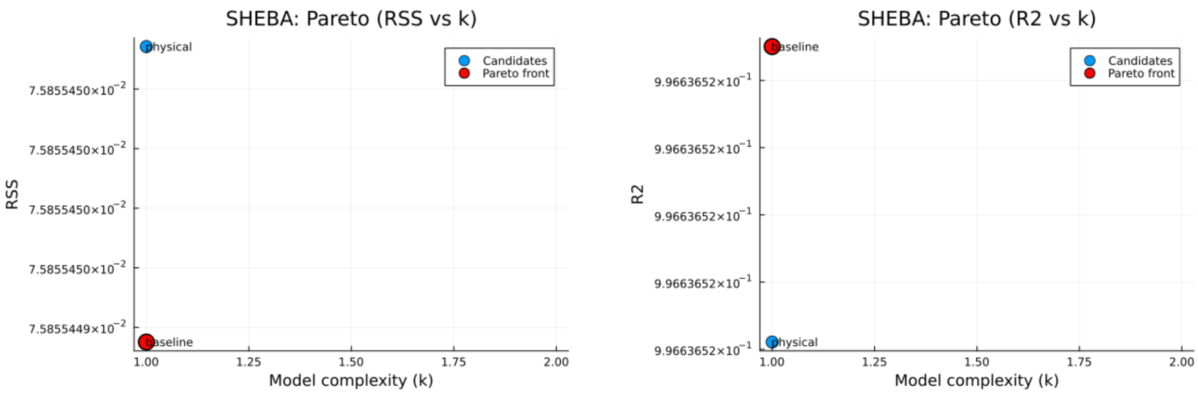
3.4 SHEBA Optimal Pareto Closure

The optimal discovered differential equation for SHEBA is:

$$\frac{\partial u}{\partial t} = 3.94889\,\frac{\partial^2 u}{\partial z^2}$$

Table 5: Discovered library terms and estimated coefficients for SHEBA.

Term	Basis	Coefficient
1	$\frac{\partial^2 u}{\partial z^2}$	3.94889
$R^2 = 0.996637$		
Residual norm $= 0.275419$		



(a) SHEBA Complexity vs RSS

(b) SHEBA Complexity vs R^2

Figure 7: Pareto trade-off fronts between closure sparsity (k) and residual metrics for SHEBA.

3.5 Spectral Galerkin Modal Profiles

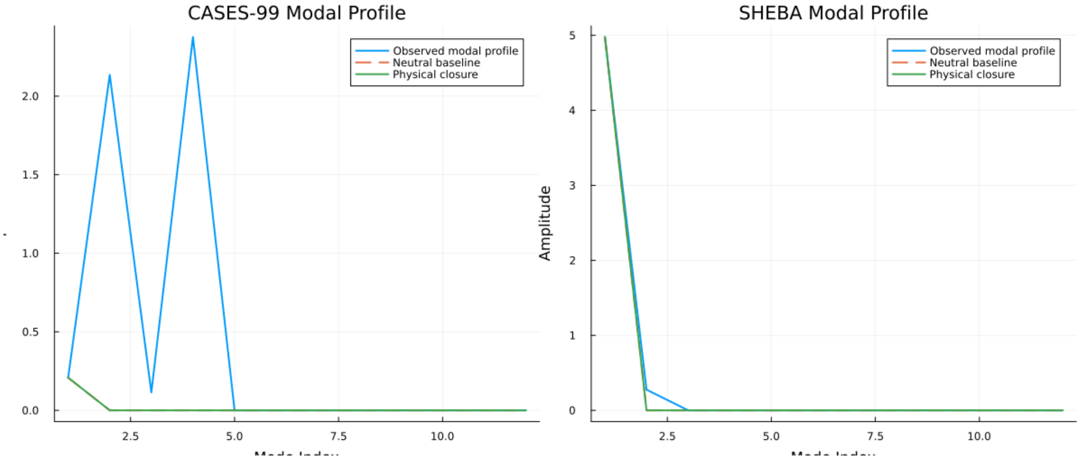


Figure 8: End-state modal profile amplitudes comparing observed mean initial states, neutral baseline predictions, and physical similarity closures.